

DAWN OF 3D TECHNOLOGY AND ITS CONSTANT EVOLUTION



When James Cameron brought *Avatar* to reality, no one dreamt about its grand success in the film industry. A few months later, color TVs, HD TVs and now FULL HD 3D TVs have begun a race for dominance. As the competition gets hotter, the sophisticated technology is becoming much more affordable for the common man. To be precise one has to pay just about 25% more for the 3D counterparts than the Flat Panel HD Displays, as per the latest estimates. The special 3D Eyewear from Panasonic is said to cost just ₹9,000. With all due credit to the cheaper and efficient Plasma panel technology, Panasonic joins the bandwagon of 3D TV era with a bang, by unleashing a wide-array of FULL HD 3D TVs into the Indian market. Two Panasonic 3D Full HD TVs (TH-P65VT20D and TH-P50VT20D) are priced at ₹3,79,000 and ₹1,99,000, respectively.

There are three variants to 3D technology – Line by line method, Checkered sampling method and Frame Sequential technology. The line by line method was used to draw stereoscopic images (left and right) sequentially one below the other, by dividing the screen into lines leading to blurred after-images while retracing a new image.

In the checkered sampling method, the screen was divided into a dot matrix, with

stereoscopic images laid out as pixels in a checkered pattern. The result was compromise in the pixel density of the final output, wherein the 3D image resolution nose-dived to half the resolution of the original HD image, leading to loss of image clarity and out of focus issues.

Frame sequential technology merges two stereoscopic images with the use of specially designed 3D Eyewear allowing



TH-P65VT20D (₹3,79,000) and
TH-P50VT20D (₹1,99,000)

optimum light to pass through the eyes and also retain the light intensity until the next image is boosted.

Panasonic cleverly adopted the Frame Sequential technology into its new breed of FULL HD 3D TVs, that promise to capture your imagination and take you closer to the virtual realism of 3D world!

The "Advanced technologies", as termed

by Panasonic are as follows:

NeoPDP 3D: It has much higher refresh rate than normal LCD displays and can contribute to higher luminescence levels without banding. The result is crisp and high-clarity images even across the borders and edges.

Quadruple luminous efficiency technology: It contributes to boost the images on the plasma screen to full brightness in 1/4th the time. This process eliminates the lack of brightness in dark scenes or night scenes that could result in loss of image clarity, by constantly refreshing the screen brightness.

Eyespeed 3D drive technology: In order to adopt the frame sequential technology of FULL HD 3D, each image has to be processed in under 1/120th of a second. Hence, in order to maintain the luminescence for such a fast processing, the luminescence time had to be reduced which could lead to lower brightness levels. So, the luminescence time was reduced by increasing the luminescence intensity at periodic intervals to maintain a uniform brightness level throughout.

Cross-talk Reduction Technology: To eliminate the major drawback inherent in conventional 3D systems, Panasonic implemented two new technologies called, "fast-decay R and G phosphors", which reduce the afterglow effects to 1/3. Thus, leading to crisp and clear 3D images sans the afterglow effects.

With all due credits, Panasonic is rightly headed towards the road to the future! On a positive note, let's hope that every household is blessed to embrace the might of 3D technology in its full glory...

